NNI-Certified Structural-Entropy Hierarchies: Fast Refinement beyond Greedy Construction

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Abstract. Structural entropy assigns an information-theoretic objective to an encoding tree, but existing hierarchy constructors need not be locally optimal under tree-topology changes. We study rooted nearest-neighbor interchange as an exact refinement operator for structural-entropy hierarchies. For weighted graphs, we derive a local cost formula involving only module volumes and two cross-module edge weights, enabling independently verified monotone descent. We incorporate this operator into SELib's multi-start hierarchical optimizer and plan a bounded compound-move extension for addressing strict-descent traps; that extension remains an evidence-gated experimental target. This is an evidence-gated development draft: final comparative and scalability claims will be inserted only from the frozen benchmark artifacts.

Keywords: Structural entropy · hierarchical clustering · local search · nearest-neighbor interchange · graph algorithms

1 Introduction

Hierarchical graph clustering seeks a tree rather than a single flat partition. Structural entropy supplies an objective for this tree [3], while recent algorithms construct binary or non-binary hierarchies through merging, partitioning, stretching, and compression [4]. Construction quality, however, does not certify local optimality in tree space.

This paper develops an exact NNI refinement layer for structural entropy. Its algorithmic contribution is separate from the companion analysis of clique landscapes: here the graph is arbitrary and weighted, the objective is the actual tree structural entropy, and the target is a reusable optimizer.

Planned contributions. The evidence plan targets (i) an exact weighted NNI delta, (ii) a monotone NNI-certified extension of a multi-start SE hierarchy optimizer, (iii) a bounded escape rule for compound moves, and (iv) paired evaluation against HCSE, BBM, Paris, and the unmodified optimizer.

2 Structural-entropy hierarchy optimization

Let $G = (V, E, w)$ be an undirected weighted graph and let an encoding-tree node a represent leaves V_a, volume $V_a = \mathrm{vol}(V_a)$, and boundary weight g_a. The tree structural entropy is

$$H^T(G) = -\sum_{a \neq r} \frac{g_a}{\mathrm{vol}(V)} \log_2 \frac{V_a}{V_{a^-}}. \tag{1}$$

The baseline optimizer combines several hierarchy initializations with strictly decreasing collapse and subtree-relocation moves. We retain every baseline candidate and add a topology-preserving binary refinement.

3 Exact NNI refinement

Consider $((A, B), C) \to (A, (B, C))$, let $P = A \cup B \cup C$, and write $W(X, Y)$ for total edge weight between modules. Only A–B and B–C edges change their lowest common ancestor, giving

$$\Delta H^T = \frac{2}{\mathrm{vol}(V)} \left[W(A, B) \log_2 \frac{V_P}{V_{A \cup B}} + W(B, C) \log_2 \frac{V_{B \cup C}}{V_P} \right]. \tag{2}$$

The implementation accepts a move only when Eq. (2) is negative and an independent full recomputation agrees with the prediction.

4 Experiments

The final evaluation will use frozen development and held-out graph seeds. It will report exact H^T, local-optimality gaps, hierarchy recovery, recovered depth, runtime, and peak memory. Every table entry will trace to a raw JSON record; the current 25-run pilot is excluded from final statistical claims.

5 Related work

Structural entropy originates with Li and Pan[3]. Pan et al. propose BBM and the HCSE stretch–compress framework[4]; Paris provides a strong graph-hierarchy initialization[1]. NNI is classical in tree space and has recently been used for hierarchical-clustering local search under a different objective[2].

6 Conclusion

This draft establishes the method and evidence boundary. The final conclusion will be written after the held-out operator and scalability benchmarks.

References

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